

# plot\_variance\_component\_triangle\_legend.R

September 10, 2026

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| plot_variance_component_triangle_legend        |
| <i>Plot Variance Component Triangle Legend</i> |

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## Description

Draws a ternary colour legend for three variance components. Each triangle corner represents the maximum value for one component, and interior points are coloured by the blended component shares.

## Usage

```
plot_variance_component_triangle_legend(  
  vec_component_colours,  
  vec_required_components = base::c("Abiotic", "Spatial", "Associations"),  
  vec_component_labels = vec_required_components,  
  max_component_value = 100,  
  component_step = 2,  
  plot_title = NULL,  
  font_family = NULL,  
  label_colour = "grey35",  
  title_colour = label_colour,  
  border_colour = "grey35",  
  background_colour = "transparent",  
  point_size = 2.3,  
  label_size = 3.2,  
  title_size = 10,  
  triangle_x_offset = 0.16,  
  method = base::c("perc_avg", "HCL")  
)
```

**Arguments**

|                                      |                                                                                                                                                                                                                                                                                                        |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>vec_component_colours</code>   | Named character vector mapping component IDs to HEX colours.                                                                                                                                                                                                                                           |
| <code>vec_required_components</code> | Character vector with exactly three component IDs. These IDs are used to read colours from <code>'vec_component_colours'</code> .                                                                                                                                                                      |
| <code>vec_component_labels</code>    | Character vector with exactly three display labels. Defaults to <code>'vec_required_components'</code> .                                                                                                                                                                                               |
| <code>max_component_value</code>     | Numeric scalar giving the value represented at each triangle corner. Defaults to <code>'100'</code> .                                                                                                                                                                                                  |
| <code>component_step</code>          | Numeric scalar giving the grid step used to fill the triangle.                                                                                                                                                                                                                                         |
| <code>plot_title</code>              | Optional character scalar shown as the plot title.                                                                                                                                                                                                                                                     |
| <code>font_family</code>             | Optional character scalar giving the font family for labels.                                                                                                                                                                                                                                           |
| <code>label_colour</code>            | Character vector giving the corner-label colour. Must have length <code>'1'</code> for one shared colour or length <code>'3'</code> for one colour per label. If length <code>'3'</code> and named, names are matched to <code>'vec_required_components'</code> ; otherwise colours are used in order. |
| <code>title_colour</code>            | Character scalar giving the title colour.                                                                                                                                                                                                                                                              |
| <code>border_colour</code>           | Character scalar giving the triangle-border colour.                                                                                                                                                                                                                                                    |
| <code>background_colour</code>       | Character scalar giving the plot background colour.                                                                                                                                                                                                                                                    |
| <code>point_size</code>              | Numeric scalar giving the size of interior colour points.                                                                                                                                                                                                                                              |
| <code>label_size</code>              | Numeric scalar giving the size of corner labels.                                                                                                                                                                                                                                                       |
| <code>title_size</code>              | Numeric scalar giving the size of the optional title.                                                                                                                                                                                                                                                  |
| <code>triangle_x_offset</code>       | Numeric scalar shifting the triangle horizontally.                                                                                                                                                                                                                                                     |
| <code>method</code>                  | Character string selecting the colour-mixing method passed to <code>[compute_variance_component_colour</code>                                                                                                                                                                                          |

**Value**

A `'ggplot'` object.

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